

Chapter 8

Kazhdan Constant

8.1 Basics

Definition 8.1. Finite group G , $\Gamma \subseteq G$, ρ a unitary representation of G on representation space V with G -invariant inner product $\langle \cdot, \cdot \rangle$. With nonempty Γ , define

$$\kappa(G, \Gamma, \rho, \langle \cdot, \cdot \rangle) = \min_{\|v\|=1} \max_{\gamma \in \Gamma} \|\rho(\gamma)v - v\|$$

when Γ is empty, define κ as 0. Note that we do not require Γ to be symmetric or a multiset.

Remark

For G -invariant $\langle \cdot, \cdot \rangle$ w.r.t. ρ , we have $\|v\| = \|\rho(g)v\|$.

Remark

Each $\|\rho(\gamma)v - v\|$ is a continuous function of v , so the max of these continuous functions is a continuous function. Also, the unit v lies in a compact sphere, hence minimum can be achieved.

Proposition 8.2. G, Γ, ρ as above, let $\langle \cdot, \cdot \rangle, \langle \cdot, \cdot \rangle'$ be two G -invariant inner products, then

$$\kappa(G, \Gamma, \rho, \langle \cdot, \cdot \rangle) = \kappa(G, \Gamma, \rho, \langle \cdot, \cdot \rangle')$$

Remark

That is, the quantity $\kappa(G, \Gamma, \rho) = \kappa(G, \Gamma, \rho, \langle \cdot, \cdot \rangle)$ is independent from the choice of inner product. Also, note a G -invariant inner product always exists.

Remark

From remark of Def. 8.1, we know the min is achieved. In fact, for any unitary irrep ρ , exists $v \in V, \gamma \in \Gamma$ s.t. $\|\rho(\gamma)v - v\| = \kappa(G, \Gamma, \rho)$

Proof

Using $\langle \cdot, \cdot \rangle$, we can decompose V (Maschke's theorem)

$$V = V_1 \oplus \cdots \oplus V_n$$

We can do the same with $\langle \cdot, \cdot \rangle'$

$$V = V'_1 \oplus \cdots \oplus V'_m$$

But we know such decomposition is essentially unique, so we have $n = m$ and a G -invariant isomorphism ϕ s.t. $\phi(V_1) = V'_1, \dots, \phi(V_n) = V'_n$. Define inner product $\langle \cdot, \cdot \rangle''$ on V by

$$\langle v, w \rangle'' = \langle \phi(v), \phi(w) \rangle'$$

Trivial to see this is indeed an inner product and is G -invariant. Note $V = V_1 \oplus \cdots \oplus V_n$ is also an orthogonal direct sum w.r.t. $\langle \cdot, \cdot \rangle''$. But by Prop. 6.37, we know invariant inner product is unique, so

$$\langle v, w \rangle'' = C_i \langle v, w \rangle$$

whenever $v, w \in V_i$, so the map

$$v_1 + \cdots + v_n \mapsto \frac{\phi(v_1)}{\sqrt{C_1}} + \cdots + \frac{\phi(v_n)}{\sqrt{C_n}}$$

defines a bijection from $\{\|v\| = 1\}$ to $\{\|v'\| = 1\}$ (mostly because ϕ is an isomorphism, so this is like change of basis). Then

$$\begin{aligned} \|\rho(\gamma) \cdot \sum_{i=1}^n v_i - \sum_{i=1}^n v_i\|^2 &= \sum_{i=1}^n \|\rho(\gamma) \cdot v_i - v_i\|^2 \\ &\quad \text{orthogonal, no intersection terms} \\ &= \sum_{i=1}^n C_i^{-1} (\|\rho(\gamma) \cdot v_i - v_i\|'')^2 \\ &= \sum_{i=1}^n C_i^{-1} (\|\rho(\gamma) \cdot \phi(v_i) - \phi(v_i)\|')^2 \\ &\quad \text{both steps by properties of inner products} \\ &= (\|\rho(\gamma) \cdot \sum_{i=1}^n \frac{\phi(v_i)}{\sqrt{C_i}} - \sum_{i=1}^n \frac{\phi(v_i)}{\sqrt{C_i}}\|')^2 \end{aligned}$$

By the bijection from v to v' , we are done.

Lemma 8.3. If ϕ and ρ are equivalent representations, then $\kappa(G, \Gamma, \phi) = \kappa(G, \Gamma, \rho)$

Proof

Let ψ be a G -isomorphism from V to W (whose existence follows from two representations being equivalent). Let $\langle \cdot, \cdot \rangle$ be a G -invariant on W , define

$$\langle v_1, v_2 \rangle' = \langle \psi(v_1), \psi(v_2) \rangle$$

whose invariance follows from G -isomorphism ψ . Note $\|v\|' = 1$ iff $\|\psi(v)\| = 1$, so

$$\begin{aligned}\kappa(G, \Gamma, \phi) &= \min_{\|v\|'=1} \max_{\gamma \in \Gamma} \|\phi(\gamma)v - v\|' \\ &= \min_{\|v\|'=1} \max_{\gamma \in \Gamma} \|\psi(\phi(\gamma)v - v)\| \\ &= \min_{\|v\|'=1} \max_{\gamma \in \Gamma} \|\rho(\gamma)\psi(v) - \psi(v)\| \\ &= \min_{\|w\|=1} \max_{\gamma \in \Gamma} \|\rho(\gamma)w - w\| = \kappa(G, \Gamma, \rho)\end{aligned}$$

The key is to use ψ as a G -invariant isomorphism.

Definition 8.4. Finite nontrivial group G , $\Gamma \subseteq G$, define

$$\kappa(G, \Gamma) = \min_{\rho} \{\kappa(G, \Gamma, \rho)\}$$

where the min is over all irreducible, nontrivial, unitary representations of G . This quantity is the Kazhdan constant.

Remark

Recall G has only finitely many irreps up to equivalence. Also, the sum of all nontrivial irreps of G is the orthogonal complement of the trivial one, so nontrivial irreps indeed exist. Hence, this min always exists.

Remark

- One possible interpretation: given any nontrivial unitary irrep ρ and any unit vector v in the representation space, there exists an element in Γ that moves v a distance of at least κ , and κ is the largest number for the preceding statement to be true.
- Another implication: Let $\rho : G \rightarrow V$ be unitary irrep, $v \in V$, then exists $\gamma \in \Gamma$ s.t.

$$\|\rho(\gamma)v - v\| \geq \kappa(G, \Gamma)\|v\|$$

- Make this a unit vector by dividing $\|v\|$
- Take $\gamma = \arg \max_{\gamma \in \Gamma} \|\rho(\gamma)w - w\|$. LHS did not take min while RHS did, done.
- Note

$$\kappa(G, \Gamma) = \min_{\rho} \min_{\|v\|=1} \max_{\gamma \in \Gamma} \|\rho(\gamma)v - v\|$$

Proposition 8.5. Finite nontrivial group G , $\Gamma \subseteq G$, then $\kappa(G, \Gamma) \leq 2$

Proof

$$\|\pi(\gamma)v - v\| \leq \|\pi(v)\| + \|v\| = 1 + 1 = 2$$

Proposition 8.6. Let $\rho : G \rightarrow GL(V)$ be a unitary representation of a finite nontrivial group G , let $\Gamma, \Gamma' \subset G$. Suppose that every element of Γ' can be expressed as a word in Γ of length less than n , then

- $\kappa(G, \Gamma, \rho) \geq \kappa(G, \Gamma', \rho)/n$
- $\kappa(G, \Gamma) \geq \kappa(G, \Gamma')/n$

Proof

(2) follows directly from (1). Let v, γ be s.t. $\kappa(G, \Gamma, \rho) = \|\rho(\gamma)v - v\| \equiv \|\gamma v - v\|$ (for notational simplicity). Let $\gamma' \in \Gamma'$, then $\gamma' = \gamma_1 \cdots \gamma_j$. Define $\alpha_k = \gamma_1 \cdots \gamma_k$ for $k = 1, \dots, j$, then

$$\begin{aligned} \|\gamma'v - v\| &= \|\alpha_j v - v\| \\ &= \|(\alpha_j v - \alpha_{j-1}v) + (\alpha_{j-1}v - \alpha_{j-2}v) + \cdots + (\alpha_2v - \alpha_1v) + (\alpha_1v - v)\| \\ &\leq \|\alpha_j v - \alpha_{j-1}v\| + \cdots + \|\alpha_1v - v\| \\ &= \|\gamma_j v - v\| + \cdots + \|\gamma_1 v - v\| \\ &\text{G-invariant} \\ &\leq j\kappa(G, \Gamma, \rho) \\ &\leq n\kappa(G, \Gamma, \rho) \end{aligned}$$

8.2 Kazhdan Constant, Isoperimetric Constant, and the Spectral Gap

Denote right regular representation as R , “the” trivial representation as 1.

Lemma 8.7. Let G be a finite nontrivial group, $\Gamma \subseteq G$, $d = |\Gamma|$, $\kappa = \kappa(G, \Gamma)$

- $\rho : G \rightarrow GL(V)$ be a nontrivial representation (i.e. $\langle \chi_1, \chi_\rho \rangle = 0$), then $\kappa(G, \Gamma, \rho) \geq \frac{\kappa}{\sqrt{d}}$
- Let $\hat{R}$ be the restriction of R to $L_0^2(G)$, $\hat{\kappa} = \kappa(G, \Gamma, \hat{R})$, then $\kappa \geq \hat{\kappa} \geq \frac{\kappa}{\sqrt{d}}$

Proof

For (2), RHS is (1) on some nontrivial irrep (which must be a part of $L_0^2(G)$), done. For LHS,

Proposition 8.8. Let G be a finite nontrivial group, $\Gamma \subseteq G$, $d = |\Gamma|$, $\kappa = \kappa(G, \Gamma)$, let $h = h(\text{Cay}(G, \Gamma))$, then

$$h \geq \frac{\kappa^2}{4d}$$

Proof

Proposition 8.9. Let G be a finite nontrivial group, $\Gamma \subseteq G$, $d = |\Gamma|$, $\kappa = \kappa(G, \Gamma)$, let λ_1 be the second largest eigenvalue, then

$$\kappa \geq \sqrt{\frac{2(d - \lambda_1)}{d}}$$

Proof

Theorem 8.10. Let d be a positive integer, let $\{G_n\}$ be a sequence of groups with $|G_n| \rightarrow \infty$. For each n , let $\Gamma_n \subseteq G_n$ s.t. $|\Gamma_n| = d$, then $\{\text{Cay}(G_n, \Gamma_n)\}$ is an expander family iff there exists $\epsilon > 0$ s.t. $\kappa(G_n, \Gamma_n) \geq \epsilon$ for all n .

Proof

With Prop.8.8 and Prop.8.9, we get a representation copy of Chapter 1.

Corollary 8.11. Let G be a finite nontrivial group, then $\kappa(G, G) > \sqrt{2}$

Proof

8.3 Abelian Groups Never Yields Expander Families

Lemma 8.12. Let θ be a real number, then $|e^{i\theta} - 1| \leq |\theta|$

Proof

By elementary calculus.

Proposition 8.13. Let G be a nontrivial finite abelian group, $\Gamma \subset G$, $d = |\Gamma|$, then

$$\kappa(G, \Gamma) \leq \frac{2\pi}{|G|^{1/d} - 1}$$

Remark

We see $\kappa \rightarrow 0$ as $|G| \rightarrow \infty$, so sequence abelian groups cannot yield expander family.

Proof

First, recall the irreps of finite abelian groups

Suppose $G = \mathbb{Z}_{n_1} \times \cdots \mathbb{Z}_{n_r}$, then the set of irreps of G is

$$\{\phi_a \mid a = (a_1, \dots, a_r) \in \mathbb{Z}_{n_1} \times \cdots \mathbb{Z}_{n_r}\}$$

where

$$\phi_a(k_1, \dots, k_r) = (e^{\frac{2\pi i a_1 k_1}{n_1}} \dots e^{\frac{2\pi i a_r k_r}{n_r}})$$

Note $|G| = |\mathbb{Z}_{n_1}| \times \cdots |\mathbb{Z}_{n_r}|$, so the size of set of all irreps is the same as $|G|$

Let $\kappa = \kappa(G, \Gamma) > 0$, C be the smallest integer greater than or equal to $2\pi/\kappa$. Claim $|G| \leq C^d$. If we have this, then $|G|^{1/d} \leq C \leq 2\pi/\kappa + 1$, solve for κ , done.

Suppose $|G| > C^d$. Let $G = \mathbb{Z}_{n_1} \times \cdots \mathbb{Z}_{n_r}$, S be the set of all irreps. We know that for any irrep ρ , for arbitrary $g \in G$, $\rho(g)$ is of the form $e^{i\theta}$ for some $\theta \in [0, 2\pi)$. Note the image of ρ will always lie inside $\mathbb{C}^*$ ($\mathbb{C}$ without 0).

For any element z on the unit circle in $\mathbb{C}$, let $\ell(z)$ be the unique argument angle s.t. $e^i = z$. Define $\phi : S \rightarrow [0, 2\pi)^d$ by

$$\phi(\rho) = [\ell(\rho(\gamma_1)), \dots, \ell(\rho(\gamma_n))]$$

where $\gamma_1, \dots, \gamma_d$ are the elements of Γ . That is, we take the argument for the mapped results of Γ

Suppose $a_1, \dots, a_d$ are integers with $1 \leq a_j \leq C$ for all j , and $a = (a_1, \dots, a_d)$, define

$$B_a = \left\{ (r_1, \dots, r_d) \in [0, 2\pi)^d : \frac{2\pi(a_j - 1)}{C} \leq r_j \leq \frac{2\pi a_j}{C} \right\}$$

That is, we partition the $[0, 2\pi)^d$. Note there are C^d choices of B_a , and the union of them all indeed partitions $[0, 2\pi)^d$. The figure below shows the case for $C = 3, d = 2$, the x-axis denotes r_1 , the y-axis denotes r_2 , and $B_{(1,2)}$ is the case $a_1 = 1, a_2 = 2$.

We know $|S| = |G|$, by the pigeonhole principle, there must be two distinct irreps ρ_1, ρ_2 falling in the same B_a box. That is, we can control their corresponding arguments

$$\omega \equiv |\ell(\rho_1(\gamma_j)) - \ell(\rho_2(\gamma_j))| < \frac{2\pi}{C}$$

for any j .

Define $\rho : G \rightarrow GL(1, \mathbb{C})$ by $\rho(g) = \rho_1(g)/\rho_2(g)$. Straightforward to see this is a homomorphism (hence a representation). Also, $\dim(GL(1, \mathbb{C})) = 1$ and $\rho_1(g)/\rho_2(g) \neq 1$, so it is a nontrivial irrep. Note $\rho(\gamma_j) = e^{i\omega}$, so by Lemma 8.12, $|\rho(\gamma_j) - 1| < 2\pi/C$ for all j . Note the unit vector in $\mathbb{C}$ is exactly 1, so $\kappa < \kappa(G, \Gamma, \rho) < 2\pi\mathbb{C}$, contradicts with definition of C , done.

8.4 Kazhdan Constant, Subgroups, and Quotients

Definition 8.14. Groups G, H , $\rho : H \rightarrow GL(V)$ a representation, $\phi : G \rightarrow H$ a homomorphism. Let $\rho^* = \rho \circ \phi$, then $\rho^* : G \rightarrow GL(V)$ is a representation of G , called the pullback of ρ via ϕ

Lemma 8.15. Finite group G , $N \triangleleft G$, suppose $\rho : G \rightarrow GL(V)$ is an irrep of G/N , let $\phi : G \rightarrow G/N$ be the canonical homomorphism (i.e. $\phi(g) = gN$), let ρ^* be the pullback of ρ via ϕ , then ρ^* is an irrep of G .

Proof

Homomorphism is trivial. Suppose there is a proper nontrivial subspace W of V s.t. $\rho^*(g)w \in W$, then $\rho(\phi(g))w \in W$, thus $\rho(gN)w \in W$, contradicting with irreducibility of ρ . Essentially, any subspace invariant under action of G is invariant under the action of G/N .

Remark

If an inner product is G/N -invariant under ρ , then it is also G -invariant under the pullback ρ^* .

Proposition 8.16. Finite nontrivial group G , finite nontrivial $N \triangleleft G$, $\Gamma \trianglelefteq G$. Let $\bar{\Gamma}$ be the image of Γ under the canonical homomorphism, then

$$\kappa(G, \Gamma) \leq \kappa(G/N, \bar{\Gamma})$$

Proof

Let $\rho : G/N \rightarrow GL(V)$ be a nontrivial irrep. Let unit vector $v \in V$, $\gamma_0 N \in \bar{\Gamma}$ be s.t. $\|\rho(\gamma_0 N)v - v\| = \kappa(G/N, \bar{\Gamma}, \rho)$ (we can do this since the min is achievable). Let ρ^* be the pullback of ρ via the canonical homomorphism, then for any $\gamma \in \Gamma$,

$$\|\rho^*(\gamma)v - v\| = \|\rho(\phi(\gamma))v - v\| = \|\rho(\gamma N)v - v\|$$

So

$$\kappa(G, \Gamma) \leq \kappa(G, \Gamma, \rho^*) = \kappa(G/N, \bar{\Gamma}, \rho)$$

this works for any ρ of G/N , done.

Remark

With Prop. 8.16 and Theorem 8.10, we recover the Quotients Nonexpansion Principle.

Proposition 8.17. Let G be a finite nontrivial subgroup, H be a nontrivial subgroup of G , $\Gamma \trianglelefteq G$, $d = |\Gamma|$, $n = [G : H]$, $T = \{t_1, \dots, t_n\}$, where T is the set of transversals for H in G . Let $\hat{\Gamma}$ be the set of Schreier generators. Then

$$\kappa(G, \Gamma) \leq d^{1/2} \kappa(H, \hat{\Gamma})$$

Proof

Recall some useful facts:

- $g \in G, g = h\bar{g}$
- $\widehat{(t, \gamma)} = t\gamma(\overline{t\gamma})^{-1}$
- Any $g \in G, g = ht$ for some unique $h \in H, t \in T$, and $t\gamma = \widehat{(t, \gamma)}\overline{t\gamma}$

Let $\sigma : H \rightarrow GL(W)$ be a nontrivial irrep of H , $\langle \cdot, \cdot \rangle$ be an H -invariant inner product on W , w be the unit vector in W s.t.

$$\kappa(H, \hat{\Gamma}) = \max_{\widehat{(t, \gamma)} \in \hat{\Gamma}} \|\sigma(\widehat{(t, \gamma)})w - w\|_W$$

Let $\rho = \text{Ind}_H^G(\sigma)$ be the induced representation, $V = \{f : G \rightarrow W \mid f(hg) = \sigma(h)f(g), \forall g \in G, h \in H\}$.

For all $h \in H, t \in T$, define

$$f(ht) = n^{-1/2}\sigma(h)w$$

Note $f(h'g) = f(h'h''t) = f(ht) = n^{-1/2}\sigma(h)w$ and $\sigma(h')f(g) = \sigma(h')f(h''t) = \sigma(h'h'')n^{-1/2}w$, so indeed $f \in V$. Also, $f(t) = n^{-1/2}w$.

Follow the inner product

$$\langle y_1, y_2 \rangle_V = \sum_j^n \langle y_1(t_1), y_2(t_1) \rangle_W$$

we have

$$\|f\|_V^2 = \sum_{j=1}^n \langle n^{-1/2}w, n^{-1/2}w \rangle_W = \sum_{j=1}^n n^{-1}\|w\|_W^2 = 1$$

so f is a unit vector in V , which can be used to get bounds for κ

Let $\gamma \in \Gamma$,

$$\begin{aligned}
\|\rho(\gamma)f - f\|_V^2 &= \sum_{j=1}^n \|f(t_j\gamma) - f(t_j)\|_W^2 \\
&\text{the definition of Ind, } \rho(x)f(g) = f(gx) \\
&= \sum_{j=1}^n \|f(\widehat{(t_j, \gamma)} t_j \gamma) - f(t_j)\|_W^2 \\
&= \sum_{j=1}^n \|\sigma(\widehat{(t_j, \gamma)}) f(\overline{t_j \gamma}) - f(t_j)\|_W^2 \\
&= \sum_{j=1}^n \|\sigma(\widehat{(t_j, \gamma)}) (n^{-1/2}w) - n^{-1/2}w\|_W^2 \\
&= \frac{1}{n} \sum_{j=1}^n \|\sigma(\widehat{(t_j, \gamma)}) w - w\|_W^2
\end{aligned}$$

Then for some j (i.e. the max of all these terms)

$$\|\rho(\gamma)f - f\|_V^2 \leq \|\sigma(\widehat{(t_j, \gamma)}) w - w\|_W^2$$

So

$$\|\rho(\gamma)f - f\|_V \leq \kappa(H, \hat{\Gamma})$$

Since γ is arbitrary, we have $\kappa(G, \Gamma, \rho) \leq \kappa(H, \hat{\Gamma})$. Also, we know from the last example of Chapter 6, ρ cannot contain the trivial representation, so by Lemma 8.7, $\kappa(G, \Gamma)/d^{1/2} \leq \kappa(G, \Gamma, \rho) \leq \kappa(H, \hat{\Gamma})$, done.

Remark

With Prop. 8.17 and Theorem 8.10, we recover the Subgroups Nonexpansion Principle.